

CAPE FEAR REGIONAL, NC, USA

WMO: 722191

Lat: 33.929N

Lon: 78.075W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.1	(c) -2.2	(d) -15.2	(e) 1.0	(f) -1.5	(g) -12.1	(h) 1.3	(i) 0.7	(j) 9.1	(k) 7.6	(l) 8.4	(m) 8.2	(n) 2.4	(o) 10	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.9	(c) 32.1	(d) 25.7	(e) 31.1	(f) 25.3	(g) 30.2	(h) 24.9	(i) 26.9	(j) 30.1	(k) 26.4	(l) 29.6	(m) 25.8	(n) 28.9	(o) 3.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	28.9	25.3	20.4	28.4	24.9	20.0	28.2	84.4	30.1	82.0	29.6	79.7	28.8	36.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.2	(c) 6.2	(e) -8.3	(f) 34.1	(g) 3.5	(h) 2.4	(i) -10.8	(j) 35.8	(k) -12.9	(l) 37.2	(m) -14.8	(n) 38.6	(o) -17.4	(p) 40.4		
(a) 8.2	(b) 7.2	(c) 6.2	(e) -9.1	(f) 28.4	(g) 1.5	(h) 2.5	(i) -10.2	(j) 30.2	(k) -11.1	(l) 31.7	(m) -11.9	(n) 33.1	(o) -13.0	(p) 34.9		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	7.5	9.2	12.3	17.0	21.4	25.3	26.4	25.8	23.8	18.7	13.0	10.1
(7)		DBStd	7.77	5.54	4.62	4.87	3.74	3.36	2.43	2.03	2.62	2.45	4.35	4.54	5.07
(8)		HDD10.0	303	118	66	33	2	0	0	0	1	0	2	20	62
(9)		HDD18.3	1343	337	257	193	66	12	1	0	3	1	49	168	257
(10)		CDD10.0	3073	39	43	105	213	353	458	509	491	413	272	109	67
(11)		CDD18.3	1071	0	1	7	27	108	208	251	234	164	61	8	3
(12)		CDH23.3	8695	0	1	26	66	581	1814	2491	2185	1199	313	18	1
(13)		CDH26.7	2204	0	0	7	3	88	447	758	629	242	29	0	0
(14)	Wind	WSAvg	2.8	2.8	2.9	3.1	3.2	2.9	3.0	3.0	2.5	2.7	2.6	2.6	2.6
(15)	Precipitation	PrecAvg	1432	107	83	103	72	96	108	151	183	201	119	83	94
(16)		PrecMax	2124	261	275	242	184	255	293	380	360	720	270	202	214
(17)		PrecMin	719	18	35	33	7	31	18	56	46	49	6	16	23
(18)		PrecStd	318	61	50	52	48	53	61	77	85	156	76	48	53
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	21.0	22.6	26.3	26.9	32.7	32.8	33.0	32.8	31.3	29.0	25.3	22.8	
(20)		0.4% MCWB	17.4	18.4	20.8	20.8	27.9	25.5	26.2	25.6	24.4	23.7	21.1	20.7	
(21)		2% DB	18.9	19.8	22.8	25.1	28.5	31.1	32.0	31.4	30.1	27.4	23.3	21.1	
(22)		2% MCWB	17.0	16.4	17.9	19.2	23.3	25.1	25.9	25.6	24.2	22.4	19.9	19.1	
(23)		5% DB	17.6	17.9	21.2	23.8	27.3	30.1	31.1	30.8	29.1	26.3	22.0	19.5	
(24)		5% MCWB	15.7	15.2	17.5	19.1	22.3	24.8	25.4	25.4	23.8	21.8	19.1	17.6	
(25)		10% DB	16.1	16.5	19.6	22.6	26.3	29.1	30.1	29.9	27.9	25.2	20.5	17.9	
(26)		10% MCWB	14.3	13.7	16.1	18.3	21.6	24.2	25.0	24.9	22.9	21.1	17.9	16.1	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	18.7	19.6	22.7	23.4	29.4	27.4	27.5	27.7	26.2	25.1	22.5	21.4	
(28)		0.4% MCDB	19.6	21.7	25.5	25.5	33.9	30.9	31.2	30.3	29.2	27.2	24.8	22.5	
(29)		2% WB	17.3	18.0	19.5	21.3	24.6	26.4	26.9	26.8	25.5	24.3	21.0	19.7	
(30)		2% MCDB	18.3	19.1	21.9	23.6	27.2	29.5	30.4	29.7	28.4	26.4	22.4	20.4	
(31)		5% WB	16.1	16.0	18.0	20.3	23.5	25.7	26.3	26.2	24.9	23.1	19.8	17.9	
(32)		5% MCDB	17.4	17.6	20.2	22.5	26.2	28.7	29.8	29.3	27.8	25.5	21.3	18.9	
(33)		10% WB	14.7	14.4	16.9	19.3	22.7	25.1	25.7	25.6	24.1	22.1	18.2	16.6	
(34)		10% MCDB	16.1	15.9	18.9	21.7	25.4	28.0	29.1	28.7	27.3	24.5	20.0	17.8	
(35)	Mean Daily Temperature Range	MDBR	9.0	9.2	9.4	8.9	7.9	6.8	6.9	7.2	7.7	9.1	10.0	9.2	
(36)		5% DB MCDBR	9.3	9.9	9.7	9.1	7.9	7.4	7.3	7.2	8.0	8.5	9.7	8.7	
(37)		5% DB MCWBR	7.8	7.0	5.9	5.3	4.2	3.7	3.4	3.4	3.7	4.8	7.0	6.8	
(38)		5% WB MCDBR	8.4	9.1	8.5	7.3	7.0	6.6	6.4	6.4	6.6	6.6	8.4	7.7	
(39)		5% WB MCWBR	7.6	7.5	6.1	4.9	4.2	4.0	3.5	3.4	3.5	4.3	7.1	6.8	
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.339	0.367	0.399	0.456	0.484	0.521	0.499	0.428	0.387	0.351	0.335	
(41)		taud	2.517	2.493	2.436	2.333	2.228	2.199	2.109	2.167	2.360	2.445	2.496	2.534	
(42)		Ebn at Noon	881	911	908	889	838	811	778	790	835	845	848	854	
(43)		Edh at Noon	84	95	109	126	142	146	158	147	116	99	85	79	
(44)	All-Sky Solar Radiation	RadAvg	2.70	3.40	4.54	5.75	6.32	6.52	6.16	5.53	4.65	3.94	3.00	2.40	
(45)		RadStd	0.18	0.33	0.40	0.39	0.41	0.39	0.39	0.33	0.34	0.41	0.22	0.18	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+165	+119

Nomenclature: See separate page